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Updated carbon footprint values for mineral fertilizer from different world regions

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Abstract

The production of mineral fertilizers, particularly nitrogen, contributes significantly to the carbon footprint of agricultural crops and food products. It is therefore important to use correct emission factors for fertilizer production, which represent current technology and efficiency of manufacturing specific fertilizer grades. The objective of this contribution is to provide life-cycle analysts with up-to-date carbon footprint reference data for the main fertilizer products globally.
At the last LCA Food Conference in 2016 we presented carbon footprint data for mineral fertilizers produced in Europe, Russia, US, and China with the technology baseline 2011. This dataset has now been updated and extended and is based on 2014 technology with global coverage representing ten main fertilizer production regions (Europe, North America, South America, Commonwealth of Independent States/CIS, Africa, Middle East, South-East Asia, South Asia, China, and Oceania).
The carbon footprint values were calculated with Fertilizers Europe's carbon footprint calculator tool (FE CFC). This FE CFC follows general LCA and carbon footprint rules, covers all main sources of greenhouse gases (GHG) emissions and has been reviewed by DNV GL to verify its completeness and correctness. While the European data are largely based on primary data reported by all Fertilizers Europe members for 2014, the data for the other regions are based on an expert evaluation of emissions from ammonia and nitric acid production in 2014.
Between 1990s and 2014, European CO_2e emissions per kg AN-N decreased from 8.69 to 3.42 kg or by about 60%. The reduction for UAN and urea was at 43% and 37%, respectively.

Keywords: *fertilizer production, product carbon footprint, GHG emissions*.

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1. Introduction

The production of mineral fertilizers, in particular nitrogen, is still largely based on fossil carbohydrates, e.g. natural gas or coal, to provide energy and hydrogen for ammonia production. In addition, the production of nitric acid can emit significant amounts of the potent greenhouse gas (GHG) nitrous oxide (N_2O), if no abatement measures are in place. Fertilizer production therefore contributes significantly to the total carbon footprint (CFP) of agricultural crops and food products. However, fertilizer production technologies and the use of CO_2-intensive fuels and feedstocks differ significantly between fertilizer products and production technology commonly employed in various regions of the world. It is therefore important to use correct emission factors for fertilizer production, which represent current technology and efficiency of manufacturing specific fertilizer

grades in a region. The objective of this paper is to provide life-cycle analysts with up-to-date carbon footprint reference data for the main fertilizer products and production regions. At the last LCA Food Conference in Dublin in 2016 we presented carbon footprint data for mineral fertilizers produced in Europe, Russia, US, and China with the technology baseline 2011. This dataset has now been updated and extended, and is based on 2014 technology with global coverage representing ten main fertilizer production regions (Europe, N-America, S-America, Commonwealth of Independent States/CIS, Africa, Middle East, SE-Asia, S-Asia, China, and Oceania).

2. Material and methods

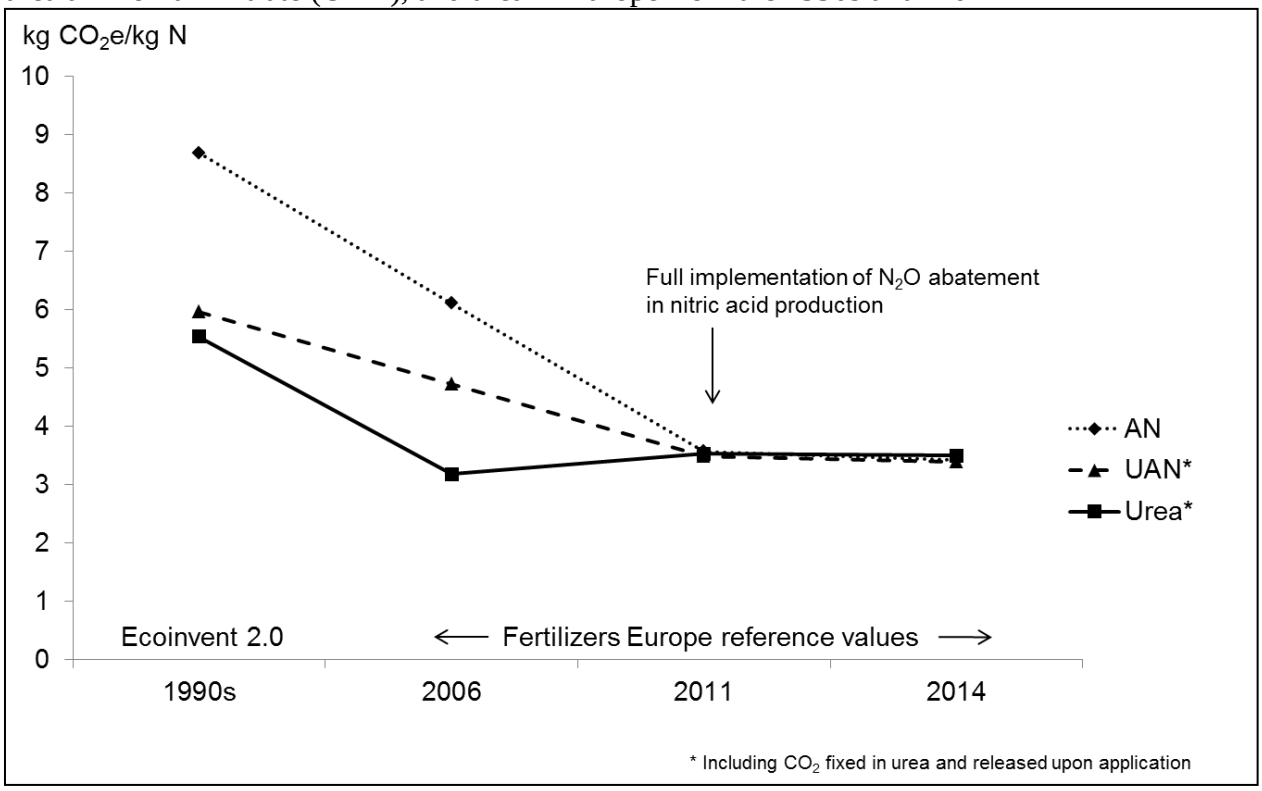
The carbon footprint values were calculated with Fertilizers Europe's carbon footprint calculator tool version 2.1 (FE CFC). The FE CFC follows general LCA and carbon footprint rules, covers all main sources of GHG emission. It has been reviewed by DNV GL to verify its completeness and correctness. While the European data are largely based on primary data reported for ammonia and nitric acid production by Fertilizers Europe members for 2014, the data for the other regions are based on an expert evaluation of emissions from ammonia and nitric acid production in 2014 (Integer Research, 2016). For all emission sources other than ammonia and nitric acid production, European default values were used in all regions. For users of the presented emission factors it is important to consider that the reference values are representative only for the respective production region. Because of the worldwide trade of fertilizer, the fertilizer product mix actually applied by farmers in a region may originate from various production regions. A specification of the tool and a description of the background data used are available from Fertilizers Europe (2016) and described in Brentrup et al. (2016).

3. Results and discussion

The FE CFC calculates the fertilizer CFPs in kg CO_2e/kg fertilizer product or kg CO_2e/kg N, where applicable. Table 1 summarizes the CFP values for all fertilizer products and all production regions expressed per kg of product. This unit was chosen since it is most commonly required as input for crop of food production LCAs. The carbon footprint values show significant differences between the production regions. For instance, the CFP value per kg of European ammonium nitrate (AN) was about one third of Chinese AN, and about half of AN produced in the other world regions. For urea the European CFP is 2.3 times the Chinese value but at the same level as in most other regions. The reason for consistently higher CFPs of Chinese N fertilizer products is that about 75% of the Chinese ammonia production is based on coal as energy and hydrogen source and only 25% on gas (Zhang et al., 2013). For all nitrate-containing products such as AN, the main differentiator is N_2O abatement technology in nitric acid production. Mainly due to political measures (EU Emission Trading System), the European fertilizer industry installed N_2O abatement technologies to avoid the cost of emission allowances.

CFP values expressed per kg of product do not allow for a comparison between products because of their different nitrogen contents. Therefore, the values need to be recalculated per kg of nitrogen by applying the nitrogen contents given in Table 1. Figure 1 shows the 2014 European CFPs of AN, liquid urea ammonium nitrate (UAN), and urea in comparison with values for the 1990s (Ecoinvent 2.0), 2006 and 2011 (both Fertilizers Europe). Between 1990s and 2014, European CO_2e emissions per kg AN-N decreased from 8.69 to 3.42 kg or by about 60%. The reduction for UAN and urea was at 43% and 37%, respectively. While for AN and partly also UAN (50% AN, 50% urea) the reduction in N_2O emissions is the main reason for the improvement, for urea and partly UAN it is improved energy efficiency that reduced the CFP values. The very low value for urea in 2006 is due to an error in energy assumption for ammonia production that was corrected for 2011 and 2014.

Figure 1: Development of carbon footprint values for production of ammonium nitrate (AN), liquid urea ammonium nitrate (UAN), and urea in Europe from the 1990s until 2014



4. Conclusions

Mineral fertilizer production often contributes significantly to carbon footprints of crop and food production systems. At the same time, fertilizer production emissions vary widely between products and production technology commonly employed in different regions of the world. Therefore, the origin of the fertilizer is important to choose the representative fertilizer production emission factors.

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Table 1: Reference carbon footprint (CFP) values for main mineral fertilizer products from different regions[8] (reference year 2014)

Fertilizer product	**Nutrient content (%)**										
		Europe	CIS (e.g. Russia)	Africa	Middle East	North America	Latin America	China	South Asia	South-East Asia	Oceania
Ammonium nitrate (gran.)[1]	33.5N	1,14	2,42	2,10	2,44	2,28	2,17	3,50	2,32	2,39	2,09
Ammonium nitrate (prill.)[1]	33.5N	1,11	2,38	2,06	2,40	2,25	2,13	3,44	2,27	2,34	2,05
Ammonium sulphate	21N, 24S	0,56	0,68	0,62	0,59	0,61	0,66	1,12	0,81	0,65	0,56
Ammonium sulphate nitrate	26N, 14S	0,80	1,39	1,22	1,33	1,29	1,27	2,08	1,42	1,35	1,18
Anhydrous ammonia[2]	82N	2,30	2,67	2,38	2,19	2,49	2,56	4,20	2,92	2,38	2,05
Calcium ammonium nitrate	27N	0,95	1,98	1,72	2,00	1,87	1,78	2,86	1,90	1,96	1,72
Calcium nitrate	15.5N	0,64	1,66	1,44	1,75	1,58	1,47	2,34	1,53	1,68	1,48
NPK (mixed-acid)[3]	15N, 15P_2O_5, 15K_2O	0,62	1,13	1,00	1,13	1,06	1,03	1,61	1,13	1,13	1,01
NPK (nitrophosphate)[4]	15N, 15P_2O_5, 15K_2O	0,71	1,22	1,09	1,22	1,15	1,11	1,71	1,24	1,22	1,10
Diammonium phosphate	18N, 46P_2O_5	0,63	0,75	0,70	0,68	0,67	0,73	1,15	0,89	0,74	0,66
Monoammonium phosphate	11N, 52P_2O_5	0,44	0,53	0,51	0,51	0,46	0,52	0,81	0,66	0,55	0,51
Super phosphate	18P_2O_5, 12S	0,08	0,09	0,10	0,11	0,08	0,09	0,13	0,13	0,11	0,11
Triple super phosphate	48P_2O_5	0,18	0,21	0,22	0,24	0,18	0,21	0,27	0,28	0,25	0,25
Urea[5]	46N	0,88	1,10	0,93	0,81	1,01	1,01	1,99	1,27	0,93	0,75
Urea ammonium nitrate (liq.)[5,6]	30N	0,78	1,43	1,23	1,34	1,33	1,29	2,20	1,44	1,36	1,17
Limestone[7]	55CaCO3	0,07	0,07	0,07	0,07	0,07	0,07	0,07	0,07	0,07	0,07
Potassium chloride[7]	60K_2O	0,25	0,25	0,25	0,25	0,25	0,25	0,25	0,25	0,25	0,25
Potassium sulphate[7]	50K_2O, 18S	0,12	0,12	0,12	0,12	0,12	0,12	0,12	0,12	0,12	0,12

[1] "gran." = granulated fertilizer product; "prill." = prilled fertilizer product
[2] Ammonia as intermediate product from Haber-Bosch process is used as a proxy for anhydrous ammonia i.e. no further processing of ammonia considered.
[3] "mixed-acid" = production of chemically blended NPK fertilizer by mixed-acid route (Fertilizers Europe, 2000a)
[4] "nitrophosphate" = production of chemically blended NPK fertilizer by nitrophosphate route (Fertilizers Europe, 2000b)
[5] For urea and urea ammonium nitrate (liq.), emissions from urea hydrolysis are not included. 0.73 kg CO_2/kg urea or 0.24 kg CO_2/kg UAN will be released after application of the product.
[6] "liq." = liquid urea ammonium nitrate solution (UAN)
[7] For these products global average default emission factors are assumed.
[8] Regions cover the following countries: Europe: EU-27. CIS (Commonwealth of Independent States): Belarus, Russia, Turkmenistan, Ukraine, Uzbekistan. Africa: Algeria, Egypt, Nigeria, South Africa. Middle East: Iran, Kuwait, Oman, Qatar, Saudi Arabia, Turkey, UAE. North America: USA, Canada. Latin America: Argentina, Brazil, Mexico, Trinidad & Tobago, Venezuela. South Asia: India, Pakistan. South East Asia: Indonesia, Malaysia, Vietnam. Oceania: Australia, New Zealand.

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